

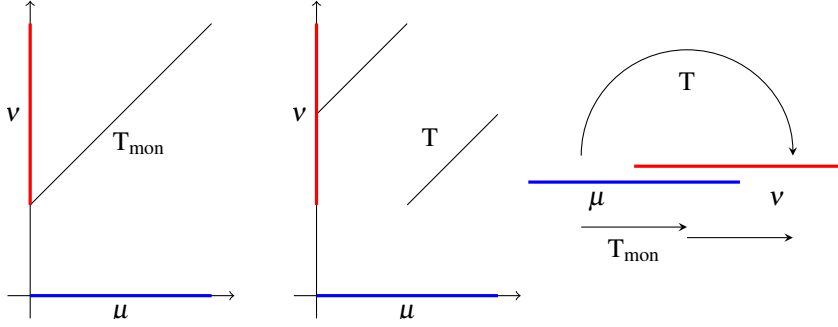


Figure 2.1: The transport maps in the book shifting example

$M(T) = \int |T_{\text{mon}}(x) - x| d\mu = 1$. Yet, the transport map T given by

$$T(x) = \begin{cases} x+2 & \text{if } x \leq 1, \\ x & \text{if } x > 1, \end{cases}$$

also satisfies $T_{\#}\mu = \nu$ and $\int |T(x) - x| d\mu = \frac{1}{2} \int_0^1 2 dx = 1$, and is optimal as well.

Starting from the fact that the optimal transport for all these costs is the monotone one we can express the cost for sending a given measure μ onto another measure ν in terms of their cumulative distribution functions F_μ and F_ν .

Proposition 2.17. *Given $\mu, \nu \in \mathcal{P}(\mathbb{R})$, consider the cost $c(x, y) = h(x - y)$, where h is a convex function. Then, we have*

$$\mathcal{T}_c(\mu, \nu) = \int_0^1 h(F_\nu^{[-1]} - F_\mu^{[-1]}) d\mathcal{L}^1.$$

If $h(z) = |z|$ then this also coincides with $\int_{\mathbb{R}} |F_\mu(t) - F_\nu(t)| dt$.

Proof. The first part of the statement is just a consequence of the optimality of γ_{mon} (Theorem 2.9).

The particular case of $h(z) = |z|$ may be treated by geometric consideration: indeed, the integral $\int_0^1 |(F_\nu)^{-1} - (F_\mu)^{-1}|$ equals the area of the part of the strip $[0, 1] \times \mathbb{R}$ bounded by the graphs of $(F_\mu)^{-1}$ and $(F_\nu)^{-1}$. In order to pass from the inverse functions to the direct ones it is enough to turn the head and look at the same strip from the variable t instead of x . But, if we want to prove it through computations, we have

$$\begin{aligned} & \int |F_\nu^{[-1]}(x) - F_\mu^{[-1]}(x)| dx \\ &= \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_\nu^{[-1]}(x) \text{ or } F_\nu^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}) \\ &= \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_\nu^{[-1]}(x)\}) \\ &\quad + \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\nu^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}). \end{aligned}$$

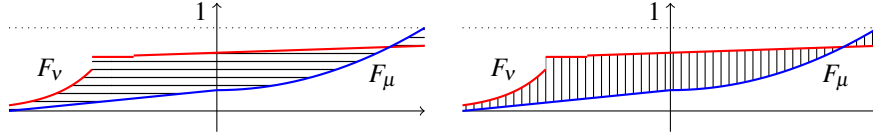


Figure 2.2: The computation of the areas in the last part of Proposition 2.17: the integral of $|F_V^{[-1]} - F_\mu^{[-1]}|$ corresponds to seeing “horizontally” the graph on the left, while, if seen “vertically”, we get the integral of $|F_\mu - F_V|$.

Then, by Fubini’s theorem, we have

$$\begin{aligned} & \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_V^{[-1]}(x)\}) \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : x \leq F_\mu(t) \text{ and } F_V(t) < x\}) dt \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : F_V(t) < x \leq F_\mu(t)\}) dt. \end{aligned}$$

Analogously,

$$\begin{aligned} & \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_V^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}) \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : F_\mu(t) < x \leq F_V(t)\}) dt. \end{aligned}$$

and, summing up,

$$\begin{aligned} & \mathcal{L}^2(\{(x, t) \in [0, 1] \times \mathbb{R} : F_\mu^{[-1]}(x) \leq t < F_V^{[-1]}(x) \text{ or } F_V^{[-1]}(x) \leq t < F_\mu^{[-1]}(x)\}) \\ &= \int_{\mathbb{R}} \mathcal{L}^1(\{x \in [0, 1] : F_V(t) < x \leq F_\mu(t) \text{ or } F_\mu(t) < x \leq F_V(t)\}) dt \\ &= \int_{\mathbb{R}} |F_\mu(t) - F_V(t)| dt. \end{aligned}$$

This concludes the proof. □ □

2.3 The Knothe transport

The Knothe transport, also known as Knothe-Rosenblatt rearrangement, is a special transport map, which has a priori nothing to do with optimal transport, that may be associated to two given measures $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$. It was independently proposed by Rosenblatt [264] for statistical purposes and by Knothe [204] for applications to geometric inequalities (see also Section 2.5.3). The main interesting point of such a map is its computational simplicity. We will explain the principle of this transport map

in the simpler case where the two measures are absolutely continuous: $\mu = f(x) dx$, $\nu = g(y) dy$.

Let us first define the densities $\hat{f}^d$ and $\hat{g}^d$ via

$$\begin{aligned}\hat{f}^d(x_d) &= \int f(t_1, t_2, \dots, t_{d-1}, x_d) dt_1 dt_2 \dots dt_{d-1}, \\ \hat{g}^d(y_d) &= \int g(s_1, s_2, \dots, s_{d-1}, y_d) ds_1 ds_2 \dots ds_{d-1}\end{aligned}$$

as well as, for $k < d$

$$\begin{aligned}\hat{f}^k(x_k, x_{k+1}, \dots, x_d) &= \int f(t_1, \dots, t_{k-1}, x_k, \dots, x_d) dt_1 \dots dt_{k-1}, \\ \hat{g}^k(y_k, y_{k+1}, \dots, y_d) &= \int g(s_1, \dots, s_{k-1}, y_k, \dots, y_d) ds_1 \dots ds_{k-1}.\end{aligned}$$

It is easy to check that $\hat{f}^d$ and $\hat{g}^d$ are the densities of $(\pi_d)_\# \mu$ and $(\pi_d)_\# \nu$, where $\pi_d : \mathbb{R}^d \rightarrow \mathbb{R}$ is the projection on the last variable. More generally, $\hat{f}^k$ and $\hat{g}^k$ are the densities of $\mu^k = (\pi_{k,d})_\# \mu$ and $\nu^k = (\pi_{k,d})_\# \nu$, where $\pi_{k,d} : \mathbb{R}^d \rightarrow \mathbb{R}^{d-k+1}$ is the map given by the projection onto the variables from k to d :

$$\pi_{k,d}(x_1, \dots, x_{k-1}, x_k, x_{k+1}, \dots, x_d) = (x_k, x_{k+1}, \dots, x_d).$$

Then, we define, for $k = 1, \dots, d-1$,

$$\begin{aligned}f^k(x_k, x_{k+1}, \dots, x_d) &= \frac{\hat{f}^k(x_k, x_{k+1}, \dots, x_d)}{\hat{f}^{k+1}(x_{k+1}, \dots, x_d)}, \\ g^k(y_k, y_{k+1}, \dots, y_d) &= \frac{\hat{g}^k(y_k, y_{k+1}, \dots, y_d)}{\hat{g}^{k+1}(y_{k+1}, \dots, y_d)},\end{aligned}$$

(note that all these functions will only be used on the set of points where the denominator does not vanish).

The function f^k , considered as a function of x_k with parameters $(x_{k+1}, \dots, x_d)$, can be actually seen as the density of the disintegration of μ^k according to the variables $(x_{k+1}, \dots, x_d)$, as well as g^k is, in the same terms, the density of the disintegration of ν^k . We briefly sketch below the main notions about the disintegration of measures. With this language, the Knothe transport that we are going to define could be defined even in the case of non-absolutely continuous measures, under some assumptions on the absence of atoms. Anyway, for the sake of simplicity, we will give most of the results in the case of absolutely continuous measures.

Box 2.2. – Important notion – Disintegrations of Measures

Definition - Consider a measure space X endowed with a Borel measure μ and a map $f : X \rightarrow Y$ valued in a topological space Y . We say that a family $(\mu_y)_{y \in Y}$ is a disintegration

of μ according to f if every μ_y is a probability measure concentrated on $f^{-1}(\{y\})$ and for every test function $\phi \in C(X)$ the map $y \mapsto \int_X \phi \, d\mu_y$ is Borel-measurable and

$$\int_X \phi \, d\mu = \int_Y dv(y) \int_X \phi \, d\mu_y,$$

where $v = f_{\#}\mu$.

The disintegration of μ is also often written (by abuse of notation) as $\mu = \mu_y \otimes v$.

In the particular case where $X = Y \times Z$ and f is the projection on the Y factor, we usually identify the measures μ_y , which are “officially” defined as measures on $Y \times Z$ concentrated on $\{y\} \times Z$, with measures on Z , so that we get

$$\int_{Y \times Z} \phi(y, z) \, d\mu(y, z) = \int_Y dv(y) \int_Z \phi(y, z) \, d\mu_y(z).$$

The disintegration of a measure μ exactly corresponds to the conditional law in probability. The reader is invited to consult [153] to find proofs about conditional probabilities and then to translate them into the disintegration language. Indeed, in probability we usually speak of the conditional law of a random variable X knowing $Y = y$. This means that the probability $\mathbb{P}$ on the probability space Ω is disintegrated according to the map $Y : \Omega \rightarrow E$ (where E is the image space of Y) into probabilities $\mathbb{P}_y$ and that we take the law of X under $\mathbb{P}_y$. The existence and the uniqueness of the disintegration depend on some assumptions on the spaces, but are true if $X = \mathbb{R}^d$.

In order to define the Knothe rearrangement, let us start from $k = d$ and define the transport map $T^d : \mathbb{R} \rightarrow \mathbb{R}$ as the monotone non-decreasing transport map sending f^d onto g^d (these functions being considered as probability densities). We know that this map is well-defined and we know how to compute it in terms of cumulative distribution functions. We will now define a family of maps $T^k : \mathbb{R}^{d-k+1} \rightarrow \mathbb{R}$, where the variables of T^k will be $(x_k, x_{k+1}, \dots, x_d)$. The first one is the map T^d that we just defined. For the sake of notations, we also define some maps $\hat{T}^k : \mathbb{R}^{d-k+1} \rightarrow \mathbb{R}^{d-k+1}$, given by

$$\hat{T}^k(x_k, x_{k+1}, \dots, x_d) = (T^k(x_k, x_{k+1}, \dots, x_d), T^{k+1}(x_{k+1}, \dots, x_d), \dots, T^d(x_d)).$$

Obviously T^d and $\hat{T}^d$ coincide. Now, if we write $f_{(x_{k+1}, \dots, x_d)}^k$ and $g_{(y_{k+1}, \dots, y_d)}^k$ for the functions $x_k \mapsto f^k(x_k, x_{k+1}, \dots, x_d)$ and $y_k \mapsto g^k(y_k, y_{k+1}, \dots, y_d)$ and we interpret them as densities of probability measures, we can define the map T^k , if we suppose that we have already defined T^j for $j > k$, in the following way: take $T_{(x_{k+1}, \dots, x_d)}^k$ to be the monotone non-decreasing transport map sending $f_{(x_{k+1}, \dots, x_d)}^k$ onto $g_{\hat{T}^{k+1}(x_{k+1}, \dots, x_d)}^k$.

Finally, the Knothe-Rosenblatt rearrangement T is defined by $T = \hat{T}^1$.

We want to check that this map T is a transport map from μ to v .

Proposition 2.18. *The Knothe-Rosenblatt rearrangement map T defined above satisfies $T_{\#}\mu = v$.*

Proof. We will prove by induction that $\hat{T}^k$ satisfies

$$\hat{T}_{\#}^k \mu^k = \nu^k \quad (2.4)$$

(let us recall that $\mu^k = (\pi_{k,d})_{\#} \mu$ and $\nu^k = (\pi_{k,d})_{\#} \nu$, are the marginals of μ and ν onto the last $d - k + 1$ variables). The equality (2.4) is fact is evident by construction for $k = d$, and if we get it for $k = 1$ we have proven $T_{\#} \mu = \nu$.

We only need to prove that, if the claim is true for a given $k + 1$, then it will be true for k . To check the equality (2.4), we just need to use test functions $\chi(y_k, \dots, y_d)$, and check

$$\int \chi(T^k(x_k, x_{k+1}, \dots, x_d), T^{k+1}(x_{k+1}, \dots, x_d), \dots, T^d(x_d)) d\mu = \int \chi(y_k, \dots, y_d) d\nu.$$

Moreover, by density (see below for Stone-Weierstrass theorem), it is enough to check the equality above on functions χ which are separable, i.e. of the form $\chi(y_k, \dots, y_d) = \phi(y_k) \psi(y_{k+1}, \dots, y_d)$. To ease the notations, for fixed k , we will denote by $\bar{x}$ the vector $\pi_{k+1,d}(x)$ and $\bar{y} = \pi_{k+1,d}(y)$. In this case we should check

$$\int \psi \circ \hat{T}^{k+1} \left(\int \phi \circ T^k f^k dx_k \right) \hat{f}^{k+1} d\bar{x} = \int \psi(\bar{y}) \left(\int \phi(y_k) g^k dy_k \right) \hat{g}^{k+1} d\bar{y}.$$

In order to get this equality, we just need to use the definition of T^k , so that we get, for every x ,

$$\int \phi \circ T^k f^k dx_k = \int \phi(y_k) g_{\hat{T}^{k+1}(\bar{x})}^k(y_k) dy_k.$$

If we define $G : \mathbb{R}^{d-k} \rightarrow \mathbb{R}$ the function given by $G(\bar{y}) = \int \phi(y_k) g_{\bar{y}}^k(y_k) dy_k$, we have now

$$\int \psi \circ \hat{T}^{k+1} \left(\int \phi \circ T^k f^k dx_k \right) \hat{f}^{k+1} d\bar{x} = \int \psi \circ \hat{T}^{k+1} G \circ \hat{T}^{k+1} \hat{f}^{k+1} d\bar{x}.$$

By taking the image measure of $\hat{f}^{k+1}$ under $\hat{T}^{k+1}$, the last expression is equal to $\int \psi(\bar{y}) G(\bar{y}) \hat{g}^{k+1}(\bar{y}) d\bar{y}$, which is in turn equal to

$$\int \psi(\bar{y}) \left(\int \phi(y_k) g_{\bar{y}}^k(y_k) dy_k \right) \hat{g}^{k+1}(\bar{y}) d\bar{y}$$

and proves the claim. $\square$ $\square$

Box 2.3. – Memo – Stone-Weierstrass Theorem

Theorem - Suppose that X is a compact space and that $E \subset C(X)$ is a subset of the space of continuous functions on X satisfying

- i) constant functions belong to E
 - ii) E is an algebra, i.e. it is stable by sum and product : $f, g \in E \Rightarrow f + g, fg \in E$
 - iii) E separates the points of X , i.e. for all $x \neq y \in X$ there is $f \in E$ such that $f(x) \neq f(y)$.
- Then E is dense in $C(X)$ for the uniform convergence (see, for instance [268]).

This theorem is usually evoked to justify the density of polynomials when $X \subset \mathbb{R}^d$. Trigonometric polynomials are also dense for the same reason (and, by the way, this may be used to prove the L^2 convergence of the Fourier series). In product spaces $X = Y \times Z$, linear combinations of separable functions of the form $\sum_i \varphi_i(y) \psi_i(z)$ are dense because of this very same theorem.

We just proved that the map T is a transport map between μ and ν . Moreover, by construction, should it be a regular map, the Knothe transport T has a triangular Jacobian matrix with nonnegative entries on its diagonal. Compare this with the Brenier optimal map for the quadratic cost: its Jacobian matrix is the Hessian matrix of a convex function, which is symmetric (instead of triangular) and has also positive eigenvalues. Moreover, we underline that assuming absolute continuity as we did, all the monotone optimal transports that we use in defining T are invertible, since not only the source measures are atomless, but also the arrival ones are atomless (they are all absolutely continuous). As a consequence, each map $\hat{T}^k$ is also invertible.

Remark 2.19. It is interesting to note that, due to its explicit definition, the Knothe map automatically inherits some regularity properties from the measures μ and ν . In general, T has the same regularity as μ and ν and not better. Indeed, each T^k is built as a monotone transport, which gives one extra derivative than the regularity of μ and ν (see Remark 2.7), but only in the direction of the variable x_k . With respect to the other variables, the explicit formula of monotone transports with cumulative distribution functions allows to prove that the dependence on $(x_{k+1}, \dots, x_d)$ is as regular as the densities are.

Remark 2.20. It is also interesting to note that the Knothe map defined above is the (unique) transport map from μ to ν which is monotone for the lexicographic order, i.e. the order where $x < y$ is defined in the following way: “there exists $k \in \{1, 2, \dots, d\}$ such that $x_j = y_j$ for $j > k$ and $x_k < y_k$ ”.

Remark 2.21. How to define the Knothe transport for μ and ν which are not absolutely continuous? The construction uses the notion of disintegration of measures (Box 2.2).

First we define the transport map $T^d : \mathbb{R} \rightarrow \mathbb{R}$ as the monotone non-decreasing transport map sending $(\pi_d)_\# \mu$ onto $(\pi_d)_\# \nu$. We need to suppose that $(\pi_d)_\# \mu$ has no atoms. Then, for each k we consider $(\pi_{k,d})_\# \mu$ and we disintegrate it with respect to $(\pi_{k+1,d})_\# \mu$, and we do the same for $(\pi_{k,d})_\# \nu$ with $(\pi_{k+1,d})_\# \nu$. We obtain some families of probabilities on the real line that we can call $\mu_{x_{k+1}, \dots, x_n}^k$ and $\nu_{y_{k+1}, \dots, y_n}^k$, respectively. They are measures on the variables x_k and y_k and they replace the densities $f^k(\cdot, x_{k+1}, \dots, x_d)$ and $g^k(\cdot, y_{k+1}, \dots, y_d)$. Again, we need to suppose that each measure $\mu_{x_{k+1}, \dots, x_n}^k$ is atomless. The maps $\hat{T}^k(x_k, x_{k+1}, \dots, x_d)$ are defined as before, using one-dimensional monotone increasing maps.

2.4 Knothe transport as a limit of quadratic optimal transports

Let us slightly modify the quadratic cost that we discussed in Section 1.3.1 and replace it with the weighted quadratic cost

$$c_\varepsilon(x, y) := \sum_{i=1}^d \lambda_i(\varepsilon) (x_i - y_i)^2$$

where the $\lambda_i(\varepsilon)$'s are positive scalars depending on a parameter $\varepsilon > 0$. If μ is absolutely continuous with respect to the Lebesgue measure, the corresponding optimal transportation problem admits a unique solution T_ε . For simplicity, in this section we will only consider the case where the supports of μ and ν are compact (but everything could be adapted to the case $\int |x|^2 d\mu, \int |y|^2 d\nu < +\infty$).

When in addition, for all $k \in \{1, \dots, d-1\}$, $\lambda_k(\varepsilon)/\lambda_{k+1}(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$, it is natural to expect the convergence of T_ε to the Knothe transport T . This convergence result was conjectured by Y. Brenier as a very natural one.

We will show it under the absolute continuity assumption of the previous section. On the other hand, [114] proves the same result under a slightly weaker assumption, namely the absence of atoms in all the disintegrated measures. The curious point is that absence of atoms is also needed on the target measure ν , and that a counter-example to the convergence result exists if it is not satisfied.

Example 2.22 (explicit computations for Gaussians). To illustrate the problem in a particular case where explicit solutions are available, take $d = 2$, and μ and ν two Gaussian measures where $\mu = N(0, I)$ and $\nu = N\left(0, \begin{pmatrix} a & b \\ b & c \end{pmatrix}\right)$ (with $ac > b^2$, $a > 0$; the notation $N(a, A)$ stands for the Gaussian density with covariance matrix A , centered at a). Take $\lambda_1(\varepsilon) = \varepsilon$ and $\lambda_2(\varepsilon) = 1$. Then it can be verified (see also Ex(11)) that T_ε is linear, and that its matrix in the canonical basis of $\mathbb{R}^2$ is

$$T_\varepsilon = \frac{1}{\sqrt{a\varepsilon^2 + c + 2\varepsilon\sqrt{ac - b^2}}} \begin{pmatrix} a\varepsilon + \sqrt{ac - b^2} & b \\ b\varepsilon & c + \varepsilon\sqrt{ac - b^2} \end{pmatrix}$$

which converges as $\varepsilon \rightarrow 0$ to $T = \begin{pmatrix} \sqrt{a - b^2/c} & b/\sqrt{c} \\ 0 & \sqrt{c} \end{pmatrix}$, i.e. the matrix of the Knothe transport from μ to ν .

We directly state our first result, whose proof, in the spirit of Γ -convergence developments (see [134] and Box 4.6), will require several steps.

Theorem 2.23. *Let μ and ν be two absolutely continuous probability measures on $\mathbb{R}^d$ with compact supports and $\gamma_\varepsilon \in \Pi(\mu, \nu)$ be an optimal transport plan for the costs $c_\varepsilon(x, y) = \sum_{i=1}^d \lambda_i(\varepsilon) (x_i - y_i)^2$, for some weights $\lambda_k(\varepsilon) > 0$. Suppose that for all $k \in \{1, \dots, d-1\}$, $\lambda_k(\varepsilon)/\lambda_{k+1}(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$. Let T be the Knothe-Rosenblatt map between μ and ν and $\gamma_K \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ the associated transport plan (i.e. $\gamma_K := (\text{id}, T)\# \mu$). Then $\gamma_\varepsilon \rightarrow \gamma_K$ as $\varepsilon \rightarrow 0$.*